Bronze data with DLT

Delta Live Tables uses Auto Loader to support schema evolution, offering significant benefits in terms of data quality, a key aspect of the Bronze layer in a Medallion architecture. In this architecture, the Bronze layer serves as the foundational stage where raw data is initially ingested and stored. DLT contributes to this layer by ensuring that each transformation applied to the raw data is precisely captured and managed. As a data scientist, understanding these transformations is crucial for maintaining the integrity of the data processing workflow. A powerful feature of DLT is its ability to automatically generate an accurate workflow **DAG** (**Directed Acyclic Graph**). This DAG not only visualizes the sequence and relationships of these data transformations but also enhances the reliability of the entire data pipeline.

The following image shows an example Delta Live Tables pipeline DAG:

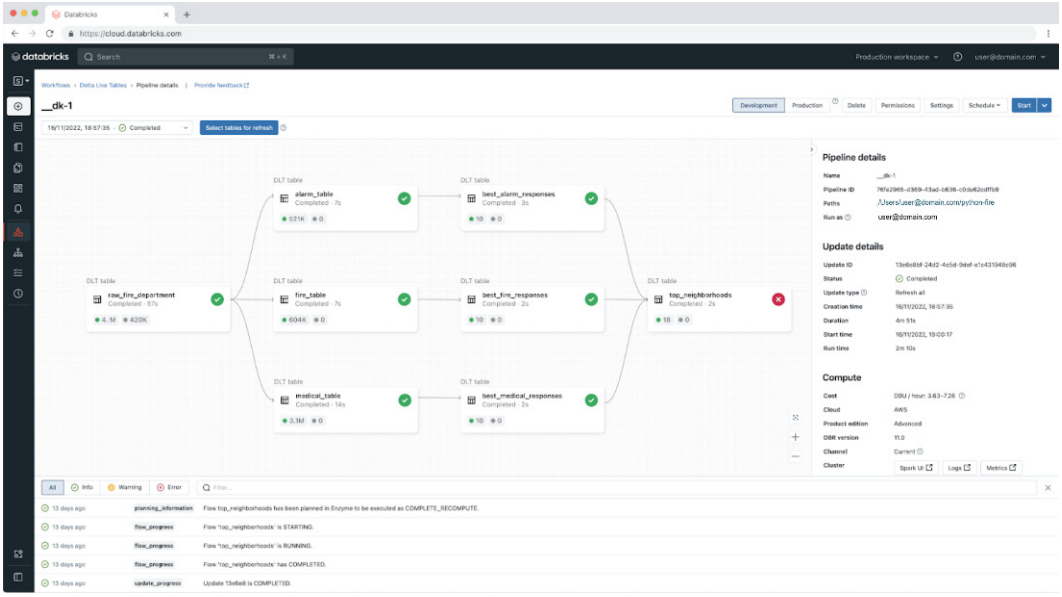


Figure 3.1 – An example Delta Live Tables pipeline workflow

Figure 3.1 is an example DLT workflow with multiple dependencies. DLT is an excellent way to build and maintain ETL pipelines, delivering high-quality data on the DI Platform. Once we have our data landed in a Delta table, there are actions we can take to ensure we are taking full advantage of Delta's benefits.

Maintaining and optimizing Delta Tables

While the primary focus of this book is not on the intricate details of Delta table optimization, understanding these techniques is crucial for building data-centric machine learning projects. Efficient management of Delta tables directly impacts the performance and reliability of ML models, as these